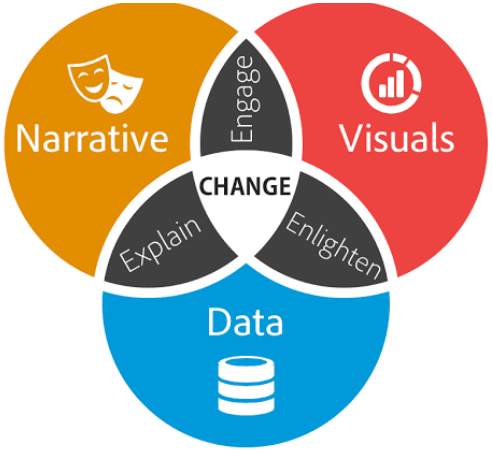
Lecture 20

Building an Effective Narrative



Announcements

- Course project deliverable 1 (exploration) now due **Monday** @ 11:59pm

- Office hours every Tuesday (noon to 4pm) and Friday (11am to 1pm)
  - Link to book appointments is on the course website

Project Info

Deliverable 1 is open ended

Goal: show me that you're familiar with the data and the scenario

- Describe and play in to your scenario
- Determine the unit of observation
- How many columns? How many numeric? How many categorical?
- Investigate some of the interesting columns: stats, plots
- Do you have missing observations? How are you handling them?
- What are some things you would like to know about the data?

Last Time

Tools for reporting our findings

- Markdown
- LaTeX (math mode)
- HTML

Today

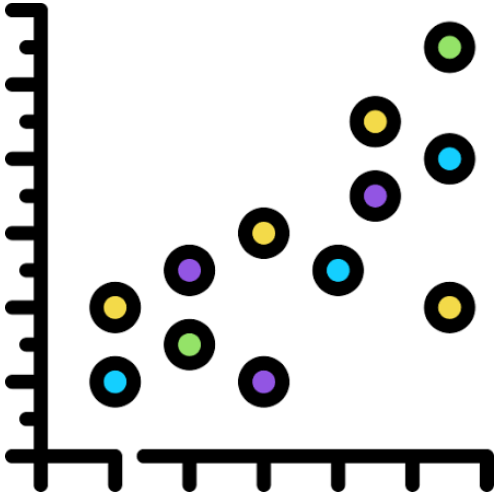
Effective communication via text

- Combining text with plots and tables
- Strategies for building a narrative
- Build on the example from last time

Reporting Tools

Plots

- Visualizations of our data
- Reveal shapes, distributions, and trends



Tables

- Displays summary statistics of our data
- Provide exact magnitudes and reference values

Text

- Provides context and guides the reader

Reporting Tools

Plots

Tables

Text

Reporting Tools

Audience most likely to read

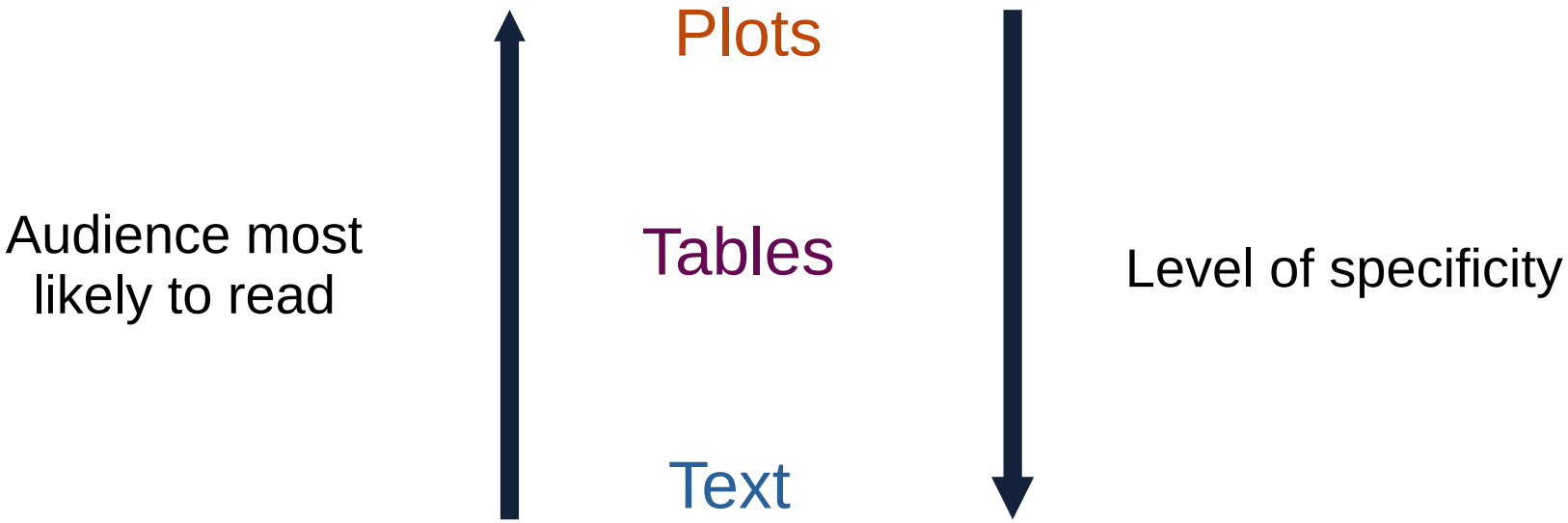


Plots

Tables

Text

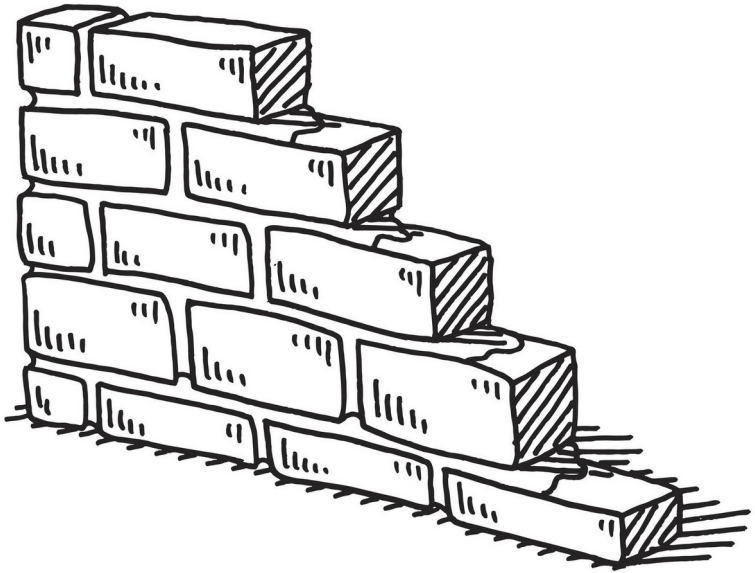
Reporting Tools



Role of Text

Text is the glue that holds your report together

- Provides context
- Explains how to interpret tables and plots
- Reinforces takeaways

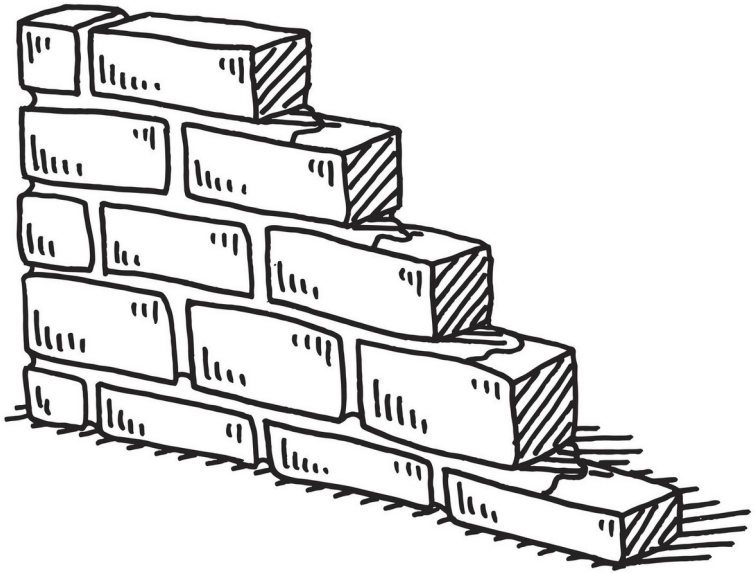


Role of Text

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- Explains how to interpret tables and plots
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Strategy: the audience should never encounter a plot or a table that is not explicitly introduced, explained, and summarized



Building an Effective Narrative

Strategy: state your main finding early on

- Don't keep the audience guessing; we're not writing a mystery novel!
- Allows you to focus your evidence on supporting findings

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Strategy: use tables and plots to complement one another

- *Weak*: "Figure 1 shows the data."
- *Strong*: "As shown in Figure 1, the distribution of running times for Algorithm A is heavily right-skewed, which agrees with the higher variance reported in Table 1."

Common Pitfalls

Causation vs. Association

- It's easy to slip into causal language when analyzing correlation
- **Strategy**: avoid words like "causes," "makes", or "determines"

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Communicating Uncertainty

- Real data is noisy; acknowledging this is not a weakness
- **Strategy**: report the spread or the presence of outliers, rather than just the mean

Exercise

- Open the lecture 19 folder

- Compare `lecture19_improved.md` and `lecture20.md`

Next Time

- Working with tables